

EE1060 Differential Equations and Transform Techniques, Exam 1

30 April 2026

Max. Marks: 20 out of 20. Time: 1 hour 50 minutes.

Instructions

- Please write and highlight the last 3 digits of your roll number prominently at the top and center of the first page of the answer sheet.
- Please also write your full name and full roll number in the space provided.
- No breaks during the exam. No laptops, calculators, mobile devices etc. allowed.
- Each step in your answer must be supported by clear mathematical reasoning.

1. (3 marks) Consider the system

$$y(t) = \int_{\tau=t}^{\infty} \frac{x(\tau)}{\tau - t + 1} d\tau.$$

Support the following questions with proof/derivation.

- Find the impulse response $h(t)$ of this system.
 - Is this system stable?
 - Is this system causal?
2. (5 marks) Find the Fourier transform of $|t|^3 u(t)$. You might find the following fact useful

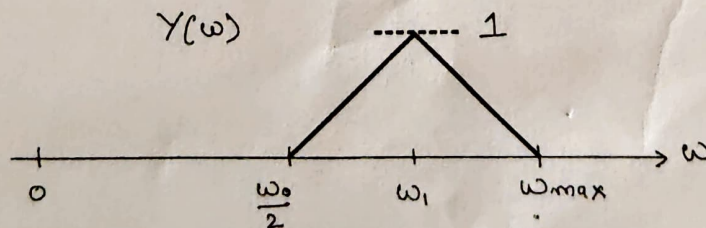
$$\text{sgn}(t) \xleftrightarrow{\mathcal{F}} \frac{2}{j\omega}.$$

3. (5 marks) The input-output relationship of a causal LTI system is

$$\frac{d^2 y(t)}{dt^2} + 2 \frac{dy(t)}{dt} + 2y(t) = x(t) + \frac{dx(t)}{dt}.$$

Use Laplace transform to find the impulse response of this system. No marks will be awarded if you use methods other than Laplace transform.

4. (7 marks) Denote the sampling interval as T_0 and let $\omega_0 = 2\pi/T_0$. Assume that $\omega_{\max} > \omega_0/2$ (i.e., the criterion for reconstruction is not satisfied) and let $Y(\omega)$ be as shown in the figure below with $Y(\omega) = 0$ for all $\omega \leq \omega_0/2$ and all $\omega \geq \omega_{\max}$. In the figure, ω_1 is the mid-point between $\omega_0/2$ and $\omega_{\max}$.
- Find $y(t)$. If needed, you can use the notation $B = \omega_{\max} - \omega_1$ in your answer.
 - Suppose $X(\omega) = Y(\omega) - Y(\omega + \omega_0)$. Sketch $X(\omega)$ and mark the coordinates carefully. Is it true that $X(\omega) = 0$ for all ω with $|\omega| \geq \omega_{\max}$?
 - Suppose $x(t)$ is sampled with sampling interval T_0 and the sampled values are used for reconstruction. Find the reconstructed signal.



Remark: Does this problem show that, in general, reconstruction is impossible when $\omega_{\max} > \omega_0/2$? No marks for answering this question.